Practical Problem 01

1.a Data description

Attribute	Description
GEO Accession	GSE58135
Disease Studied	ER+ (Estrogen Receptor-positive) Breast Cancer
Organism	Homo sapiens (human)
Tissue / Sample Type	Primary ER+ breast tumor tissue vs. uninvolved breast tissue adjacent to the ER+ tumor
Sequencing Platform	Illumina HiSeq 2000 (GEO platform GPL11154)
Case Samples (n)	42 ER+ primary tumors
Control Samples (n)	30 uninvolved breast tissue samples adjacent to ER+ tumors
Experimental Design	Expression profiling by high throughput sequencing

1.b Data acquisition, filtering and Normalization

Data base : NCBI GEO datasets

Genes with CPM of grater or equal to 1 in at least 30 samples were retained and all other genes were filtered out.

Number of genes before filtering: 39376

Number of genes after filtering: 18118

Normalization:

Differential Analysis	Normalization Technique
DESeq2	Median of ratios
EdgeR	TMM
Limma-VOOM	TMM and Log2 CPM

1.C Differential analysis

DGE analysis was performed using DESeq2, edgeR, and Limma-voom package in R. In each method, genes were called significant DEGs at adjusted p-value (FDR) < 0.05 and |log2FC| > 2.

Method	Up regulated	Down regulated	Total DEGs
DESeq2	625	720	1,345
edgeR	596	868	1,464
Limma-voom	353	822	1,175

Volcano plots

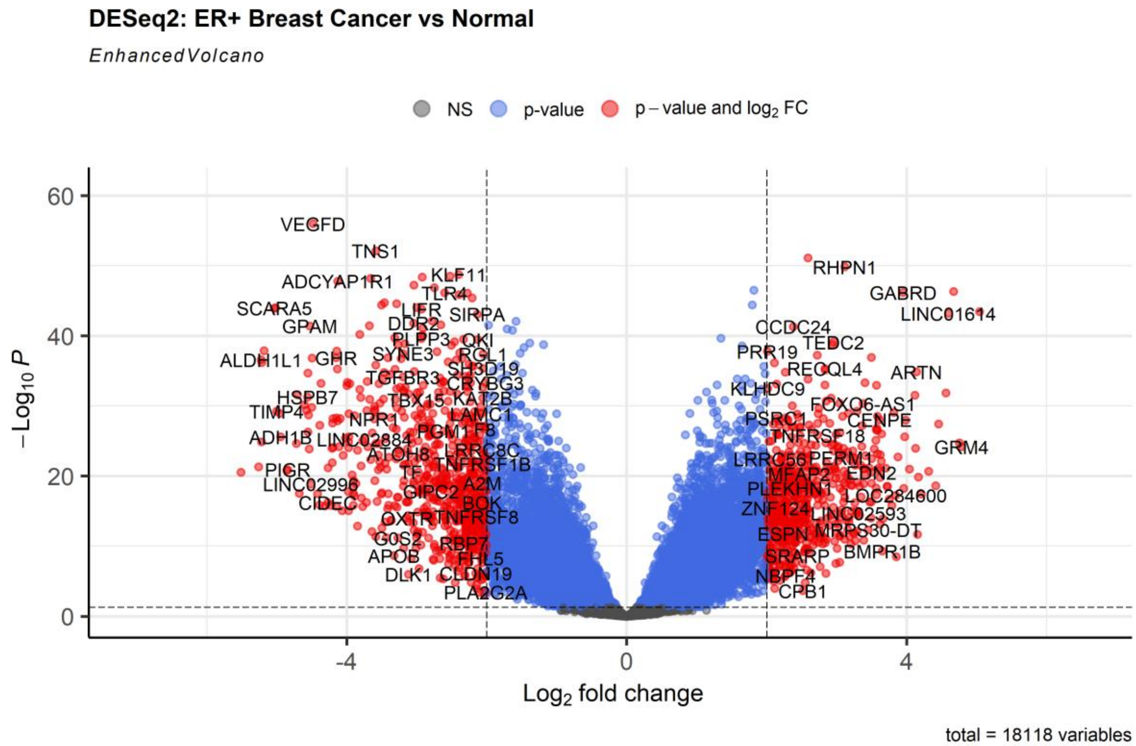

Figure 1. DESeq2 volcano plot.

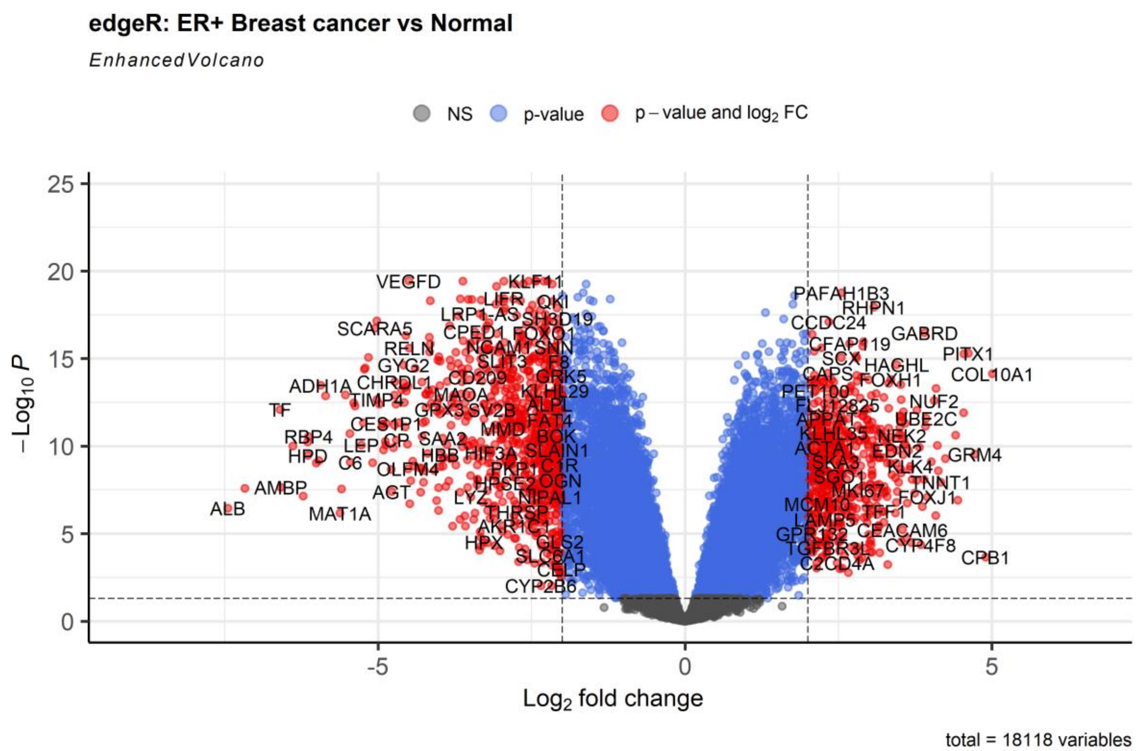

Figure 2. edgeR volcano plot.

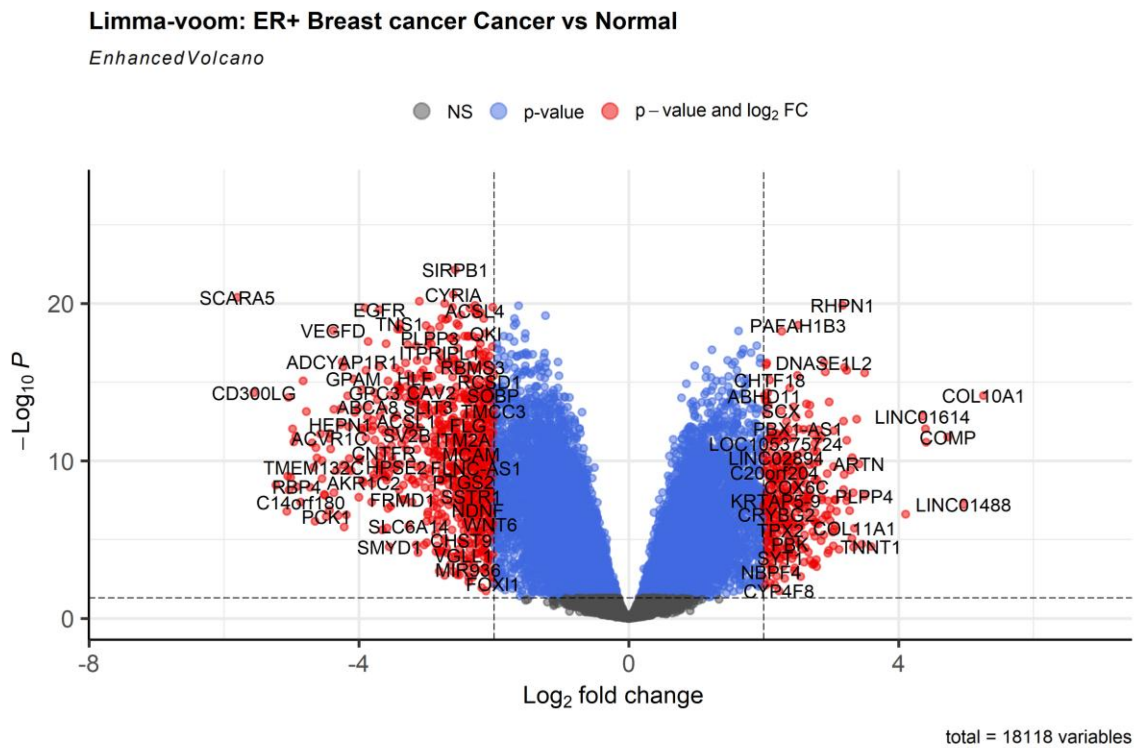

Figure 3. Limma-voom volcano plot.

Comparison

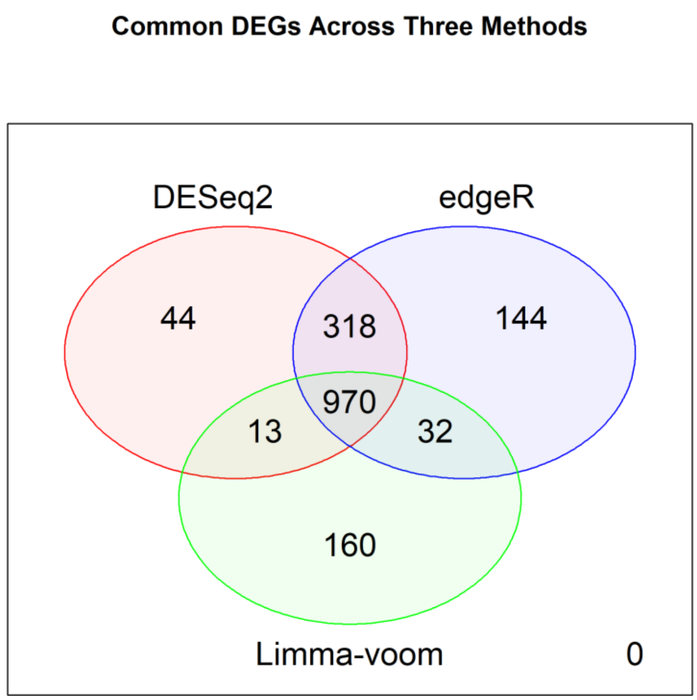

Figure 4. Venn-diagram of significant DEGs across DESeq2, edgeR, and Limma-voom.

1681 unique DEGs were identified across all three techniques. DESeq2 and edgeR produced similar totals while Limma-voom produced fewer DEGs. Limma produced most unique DEGs (160). DESeq2 produced least (44). 970 DEGs out of 1681 were common across all three techniques.

1.d Top 10 significant DEGs and therapeutic potential

The 970 DEGs common to all three methods were used to build a PPI network (NetworkAnalyst). Genes were ranked by network degree measure to obtain the ten most connected hub genes. Each gene was queried against DSigDB through enrichR.

Rank	Gene	Degree	Top DSigDB drug
1	EGFR	130	trifluridine (MCF7 DOWN)
2	H3C4	121	etoposide (MCF7 DOWN)
3	GNAI1	114	protoporphyrin IX (CTD 00001323)
4	KAT2B	80	ciclopirox (MCF7 DOWN)
5	PLK1	78	5109870 (MCF7 DOWN)
6	CDC20	77	hematoxylin (BOSS)
7	BUB1B	70	Bisphenol A (CTD 00000312)
8	H2AC18	69	Fulvestrant (CTD 00002740)
9	GNG2	64	Tyrphostin AG 1478 (TTD 00011607)
10	KDR	63	Vatalanib succinate (TTD 00011775)

Practical Problem 02

2.a Data description

Two datasets from the same GEO series (GSE58135) were used, one per breast cancer subtype, each with its own matched normal samples.

Attribute	Dataset 1: ER+	Dataset 2: Triple-Negative (TN)
GEO Accession	GSE58135	GSE58135
Disease Studied	ER+ Breast Cancer	Triple-Negative Breast Cancer
Organism	Homo sapiens	Homo sapiens
Tissue / Sample Type	Primary ER+ tumor vs. adjacent normal	Primary TN tumor vs. adjacent normal
Sequencing Platform	Illumina HiSeq 2000 (GPL11154)	Illumina HiSeq 2000 (GPL11154)
Case Samples (n)	42	42
Control Samples (n)	30	21
Experimental Design	Expression profiling by high throughput sequencing	Expression profiling by high throughput sequencing

2.b Data acquisition, filtering and Normalization

Data base : NCBI GEO datasets

Genes with CPM of grater or equal to 1 in at least 21 samples were retained and all other genes were filtered out.

Number of genes before filtering: 39376

Number of genes after filtering: 18846

Normalization:

Differential Analysis	Normalization Technique
DESeq2	Median of ratios
EdgeR	TMM
Limma-VOOM	TMM and Log2 CPM

2.C Differential analysis (each dataset)

Dataset 1: ER+ Breast Cancer

Method	Up regulated	Down regulated	Total DEGs
DESeq2	625	720	1,345
edgeR	596	868	1,464
Limma-voom	353	822	1,175

Dataset 2: Triple-Negative Breast Cancer (Criteria: FDR < 0.05, |log2FC| > 2)

Method	Total DEGs
DESeq2	1,924
edgeR	1,951
Limma-voom	1,786

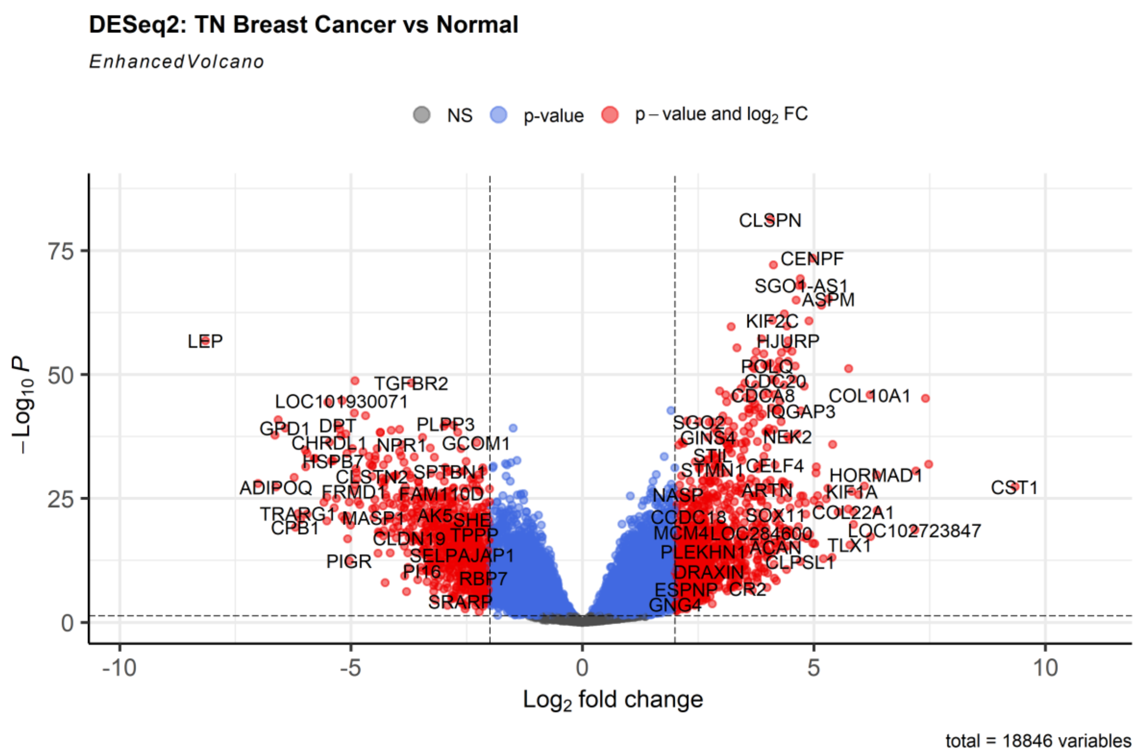

Figure 5. DESeq2 volcano plot, TN breast cancer vs. normal.

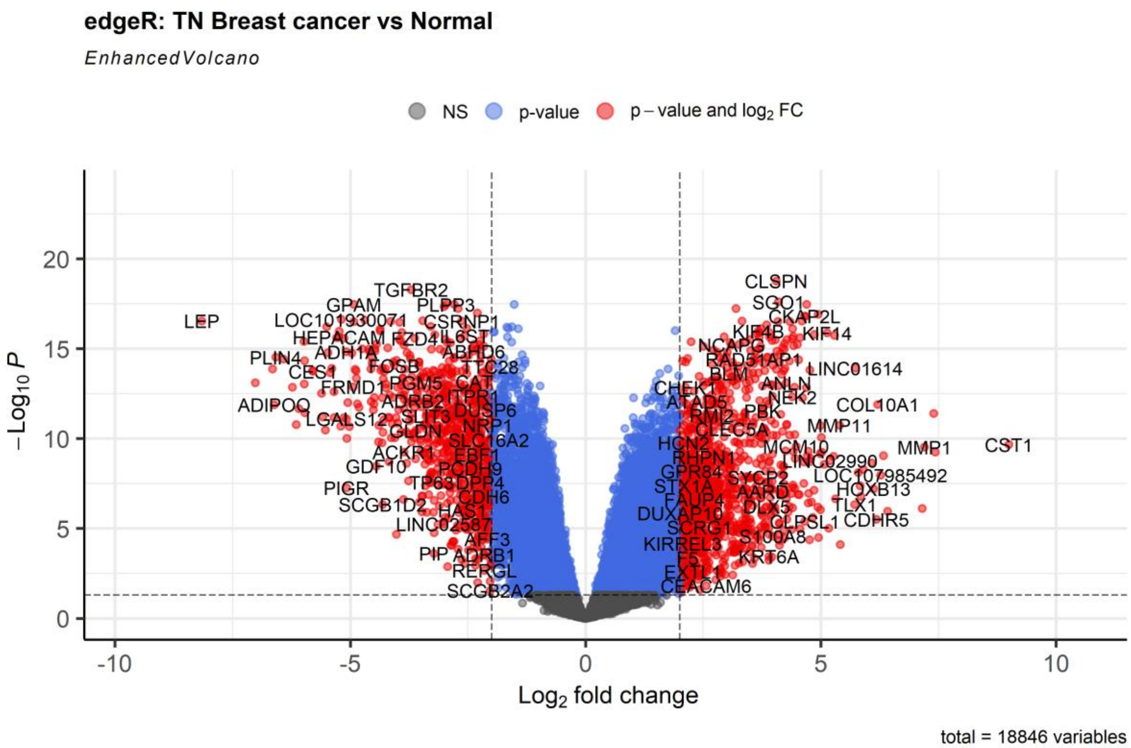

Figure 6. edgeR volcano plot, TN breast cancer vs. normal.

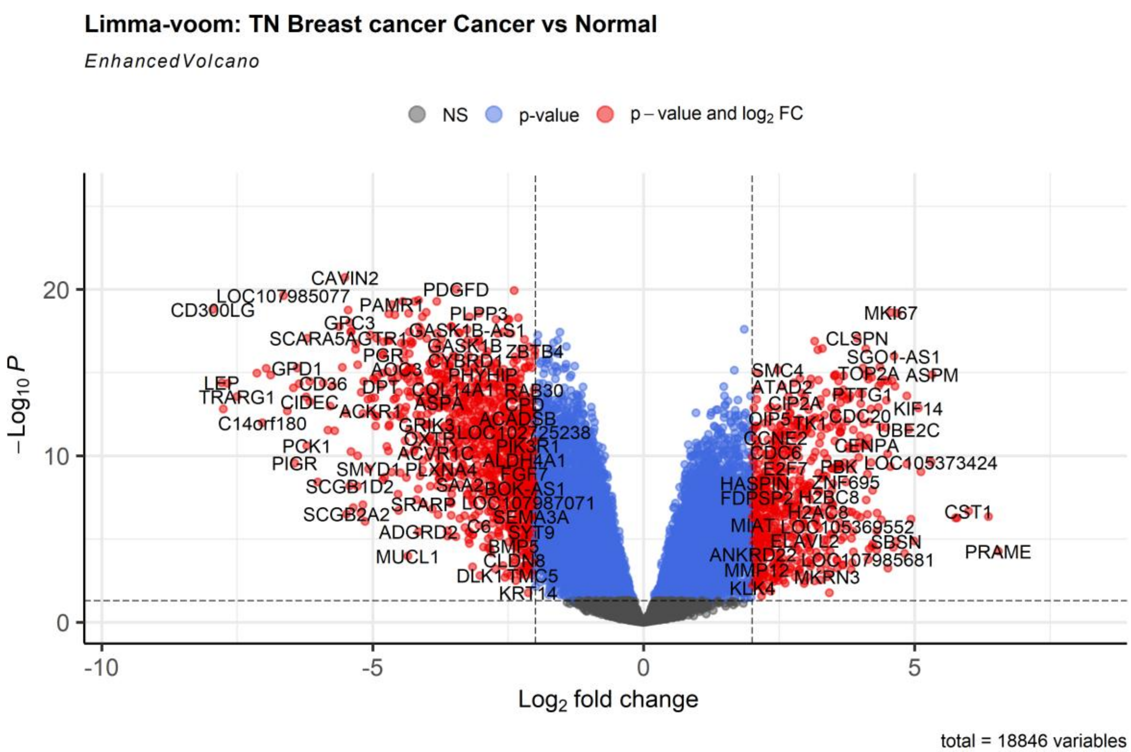

Figure 7. Limma-voom volcano plot, TN breast cancer vs. normal.

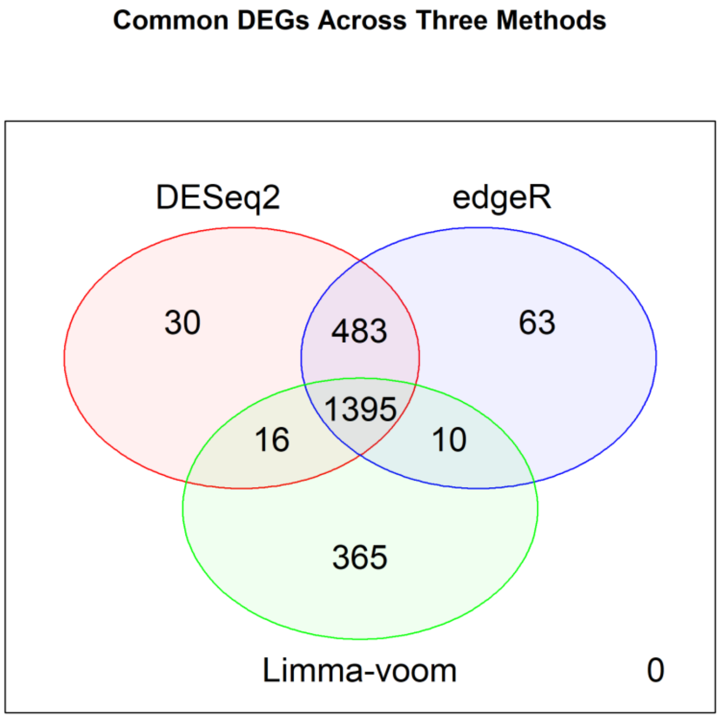

Figure 8. Venn-diagram of significant DEGs across DESeq2, edgeR, and Limma-voom, TN dataset.

2362 unique DEGs were identified across all three techniques. Limma-voom produced most unique DEGs (365). DESeq2 produced least (30). 1395 DEGs out of 2362 were common across all three techniques.

Comparison across datasets

The TN dataset produced more DEGs than the ER+ dataset at every method. In both cases Limma-voom produced the most unique DEGs. Intersecting the two common DEG sets (970 ER+ genes and 1,395 TN genes) gave 527 common DEGs for both subtypes of breast cancer.

2.d Top 10 common DEGs and therapeutic potential

The 527 common DEGs for both datasets were used to build a PPI network (NetworkAnalyst). Genes were ranked by degree to obtain the ten most connected hub genes. Each gene was queried against DSigDB through enrichR.

Rank	Gene	Degree	Top DSigDB drug/compound
1	H3C4	121	trifluridine (MCF7 DOWN)
2	GNAI1	114	etoposide (MCF7 DOWN)
3	PLK1	78	ciclopirox (MCF7 DOWN)
4	CDC20	77	5109870 (MCF7 DOWN)
5	BUB1B	70	resveratrol (MCF7 DOWN)
6	H2AC18	69	thalidomide (CTD 00006858)
7	GNG2	64	dmnq (CTD 00002569)
8	KDR	63	362-05-0 (CTD 00000041)
9	UBE2C	61	methotrexate (MCF7 DOWN)
10	FOXO1	57	caffeic acid (CTD 00001850)

Practical Problem 03

3.a Data description

Attribute	Description
Accession	PRJNA871997
Disease Studied	Type 2 Diabetes (T2D)
Organism (Host)	Homo sapiens (human)
Sample Type	Stool / gut microbiome
Sequencing Platform	Illumina MiSeq, 16S rRNA gene (V3-V4) amplicon sequencing
Case Samples (n)	10 T2D subjects
Control Samples (n)	13 healthy subjects
Experimental Design	Case-control comparison of gut bacterial community composition
Objective	To characterize gut microbiota differences between T2D patients and healthy subjects

3.b Data acquisition, filtering and Normalization

Filtered using a prevalence threshold: taxa retained if present in at least 5% of samples.

Normalization:

Analysis	Normalization Technique
Alpha / Beta diversity, Taxonomy	Rarefaction to even sequencing depth
metagenomeSeq	Cumulative Sum Scaling (CSS)
DESeq2	Median of ratios
edgeR	Library-size normalization

3.c Alpha and Beta diversity

Alpha diversity

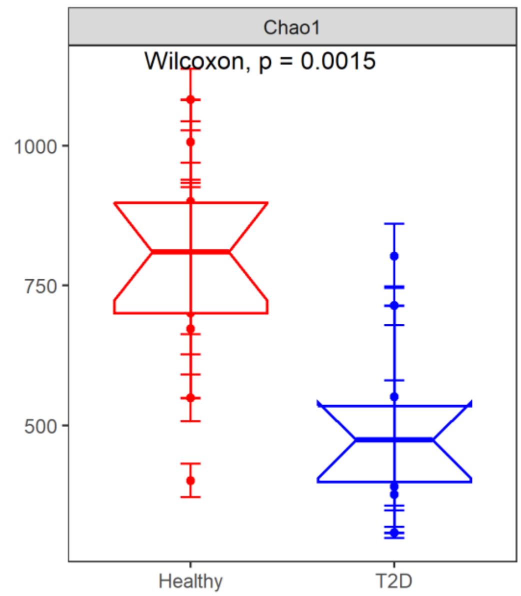

Figure 1. Chao1 alpha diversity, Healthy vs. T2D.

Chao1 richness was significantly higher in Healthy subjects than in T2D subjects (Wilcoxon, p = 0.0015), indicating reduced gut bacterial richness in T2D.

Beta diversity

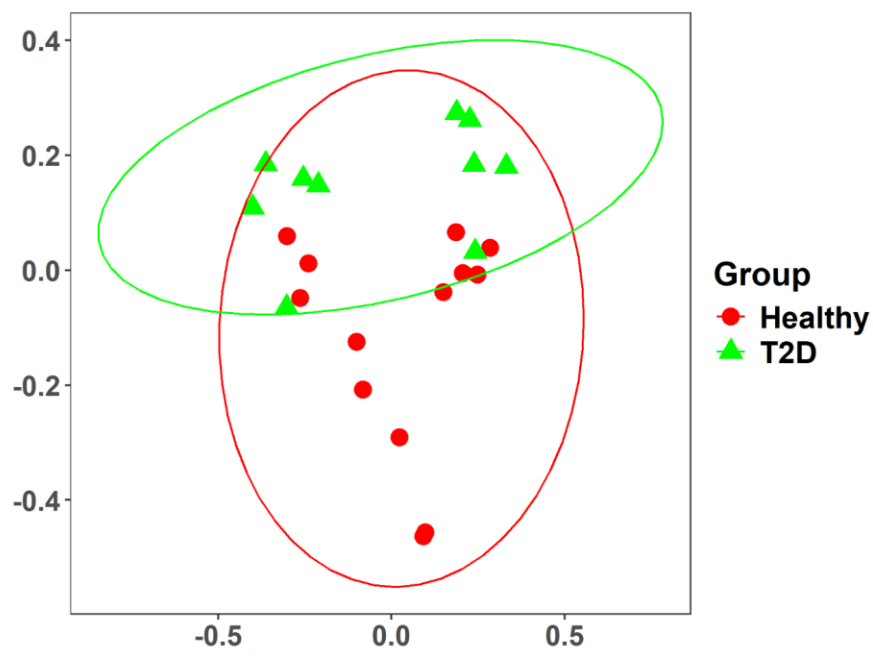

Figure 2. Bray-Curtis PCoA, Healthy vs. T2D.

PCoA plot showed separation between Healthy and T2D samples. PERMANOVA test also confirmed it (p-value = 0.047).

3.d Taxonomy analysis

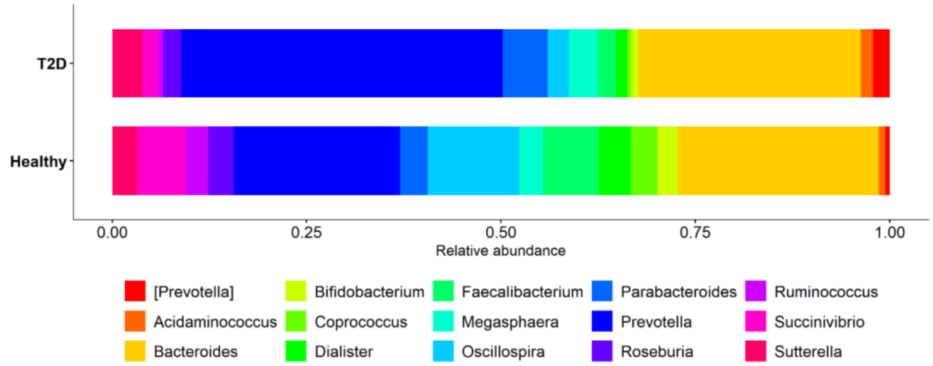

Figure 3. Top 15 genera by relative abundance, Healthy vs. T2D.

Prevotella was the dominant genus in both groups but occupied larger share of the community in T2D than in Healthy samples.

3.e Differential abundance analysis

Taxa were called significantly differentially abundant at adjusted p-value (FDR) < 0.05.

Method	Significant Taxa
metagenomeSeq	2
DESeq2	117
edgeR	140

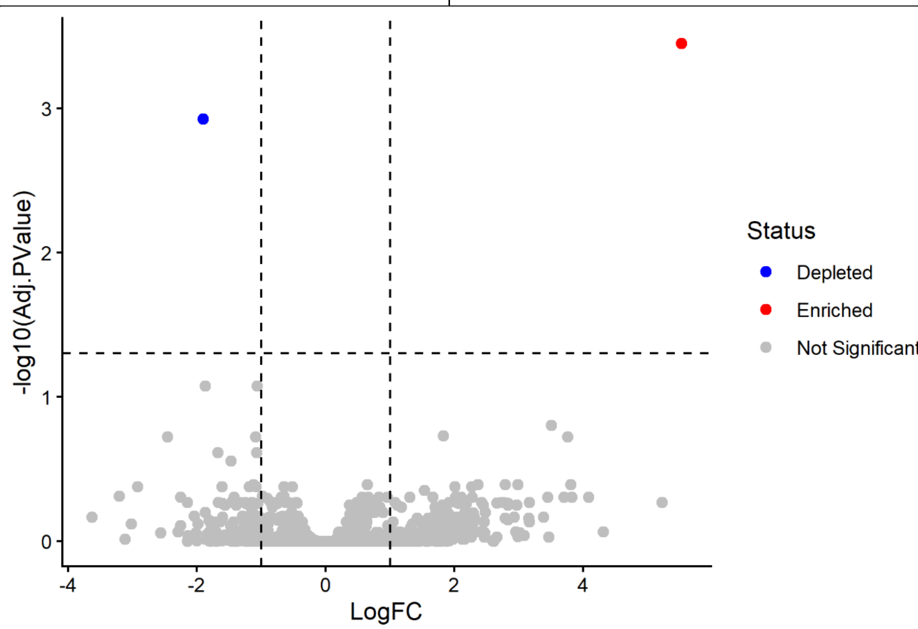

Figure 4. metagenomeSeq volcano plot.

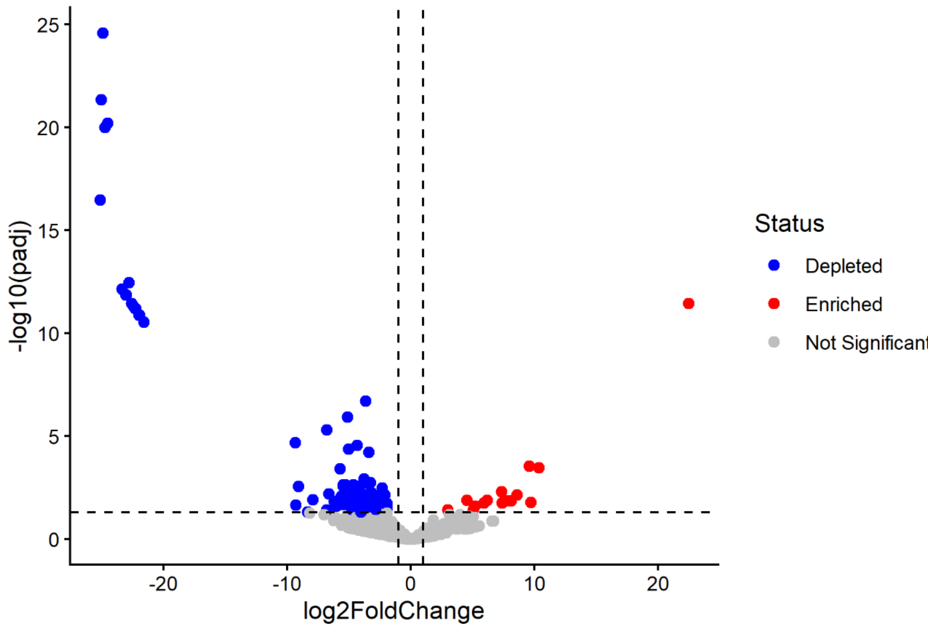

Figure 5. DESeq2 volcano plot.

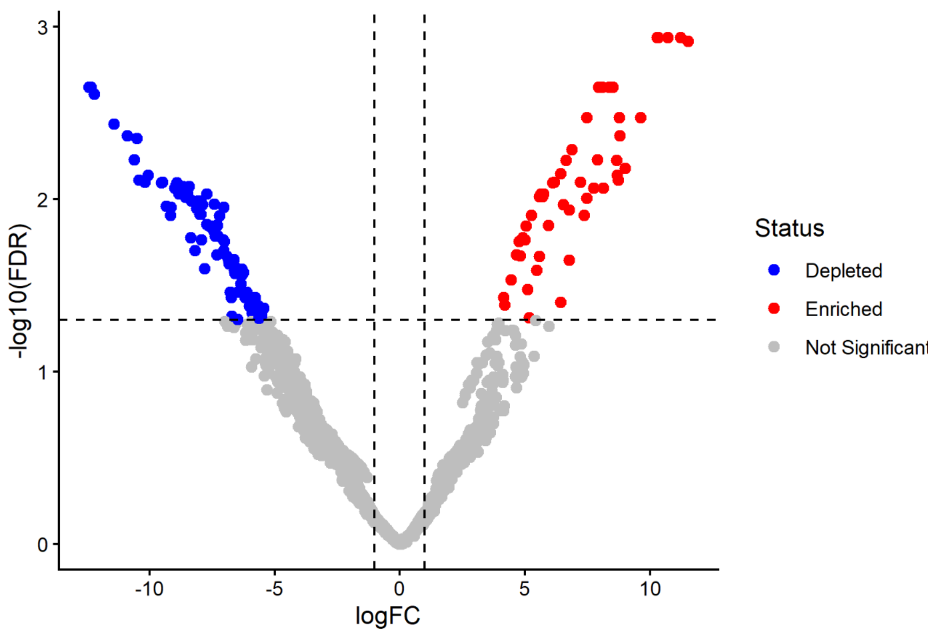

Figure 6. edgeR volcano plot.

Practical Problem 04

4.a Data description

Attribute	Dataset 1	Dataset 2
Accession	PRJNA871997	PRJNA820272
Disease Studied	Type 2 Diabetes (T2D)	Type 2 Diabetes (T2D)
Organism (Host)	Homo sapiens	Homo sapiens
Sample Type	Stool / gut microbiome	Stool / gut microbiome
Sequencing Platform	Illumina MiSeq, 16S rRNA (V3-V4)	16S rRNA gene amplicon sequencing
Experimental Design	Case-control: T2D vs. Healthy	Case-control: T2D vs. Healthy
Objective	Characterize gut microbiota differences in T2D vs. healthy subjects	Characterize gut microbiota differences associated with T2D-linked dysbiosis

4.b Data acquisition, filtering and Normalization

Both datasets were filtered independently using the same prevalence threshold as Problem 3 (taxa present in at least 5% of samples).

The same normalization techniques used in Problem 3 (Section 3.b) were applied independently to each dataset.

4.c Alpha and Beta diversity (each dataset)

Alpha diversity

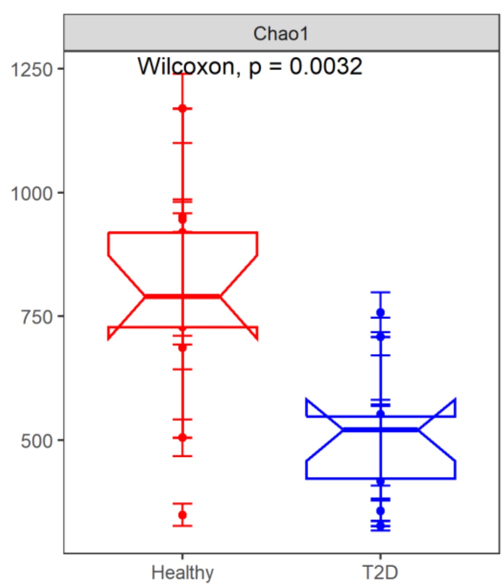

Figure 7. Chao1 alpha diversity, Dataset 1 (Healthy vs. T2D).

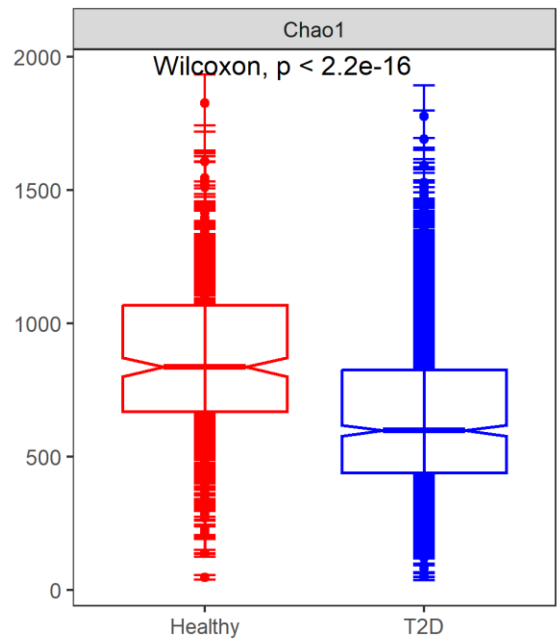

Figure 8. Chao1 alpha diversity, Dataset 2 (Healthy vs. T2D).

Both datasets showed significantly higher richness in Healthy subjects than in T2D subjects (Dataset 1: Wilcoxon p = 0.0032; Dataset 2: Wilcoxon p < 2.2e-16).

Beta diversity

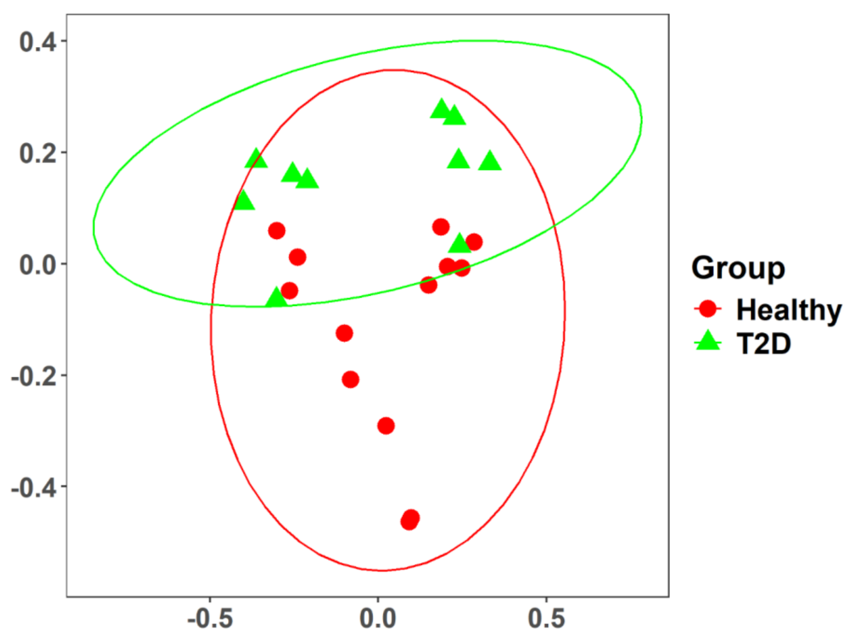

Figure 9. Bray-Curtis PCoA, Dataset 1.

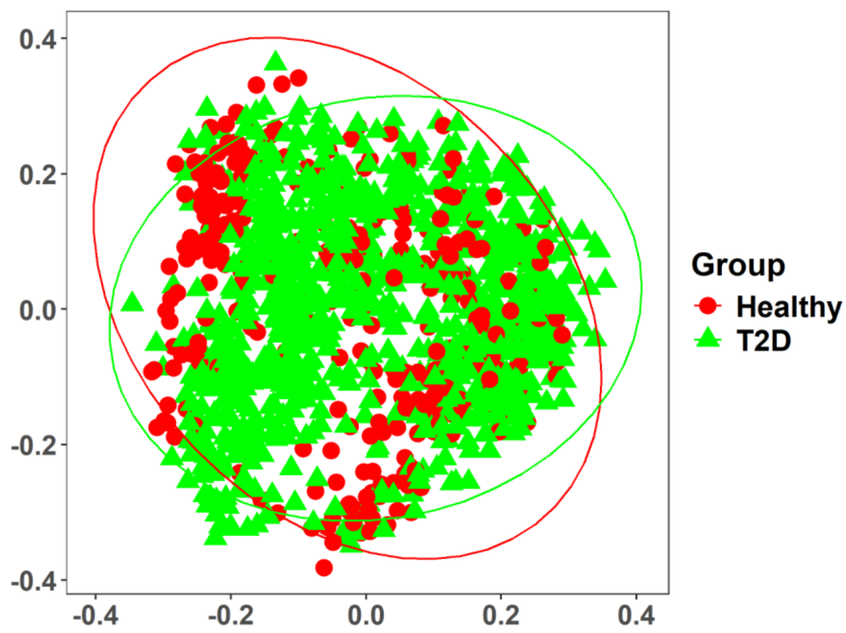

Figure 10. Bray-Curtis PCoA, Dataset 2.

Both datasets showed partial separation between Healthy and T2D samples.

4.d Taxonomic composition

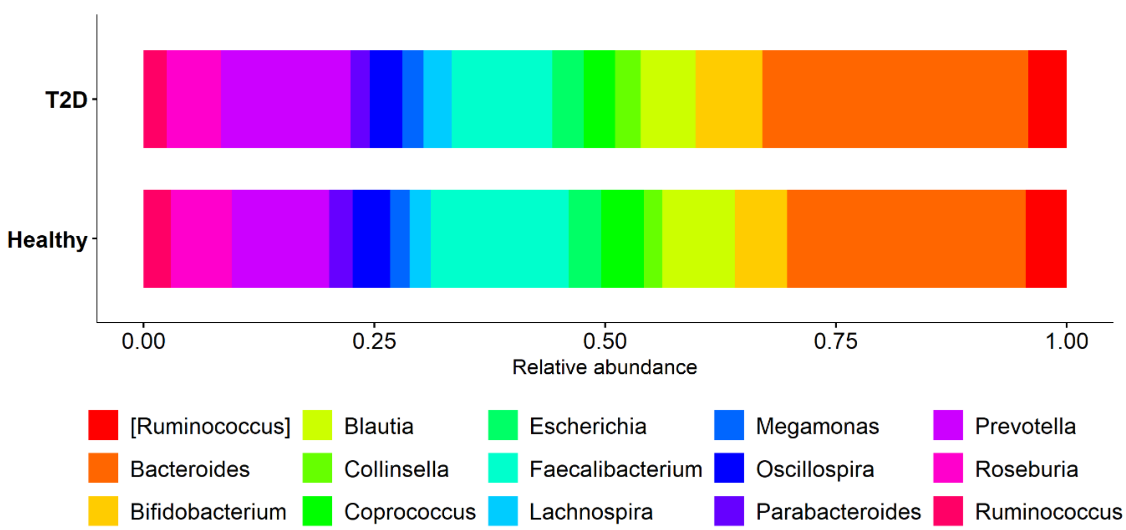

Figure 11. Top 15 genera by relative abundance, Dataset 2.

Dataset 1 was dominated by Prevotella and Bacteroides (Figure 3). Dataset 2 showed a broader representation of Firmicutes-associated genera, including Blautia, Coprococcus, Lachnospira, and Roseburia, alongside Bacteroides and Prevotella, indicating a more even genus-level composition than Dataset 1.

4.e Differential abundance analysis (each dataset)

Significance criterion: adjusted p-value (FDR) < 0.05, applied independently to each dataset.

Dataset 1

Same as Problem 3, Section 3.e: metagenomeSeq = 2, DESeq2 = 117, edgeR = 144 significant taxa (Figures 4-6).

Dataset 2

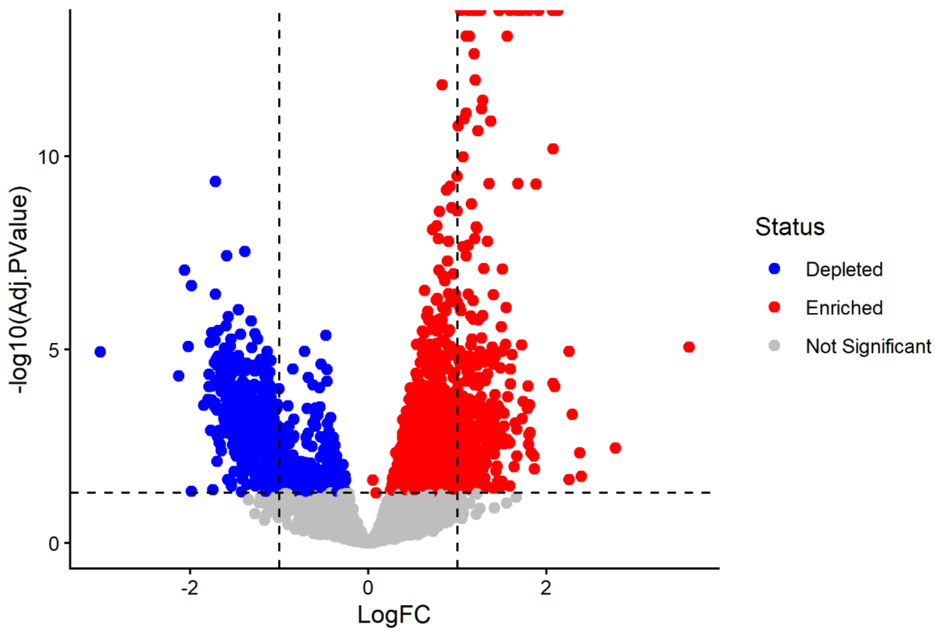

Figure 12. metagenomeSeq volcano plot, Dataset 2.

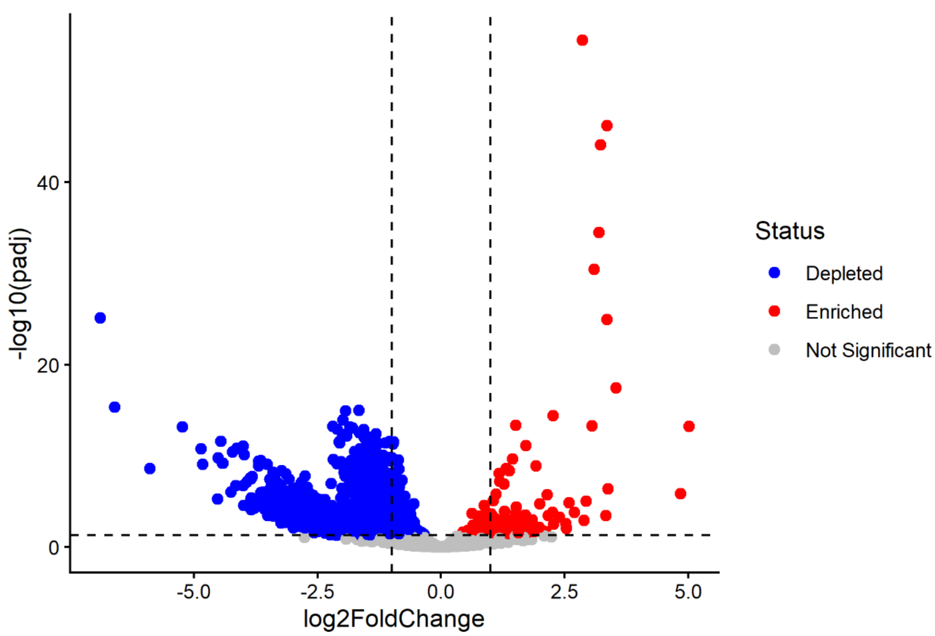

Figure 13. DESeq2 volcano plot, Dataset 2.

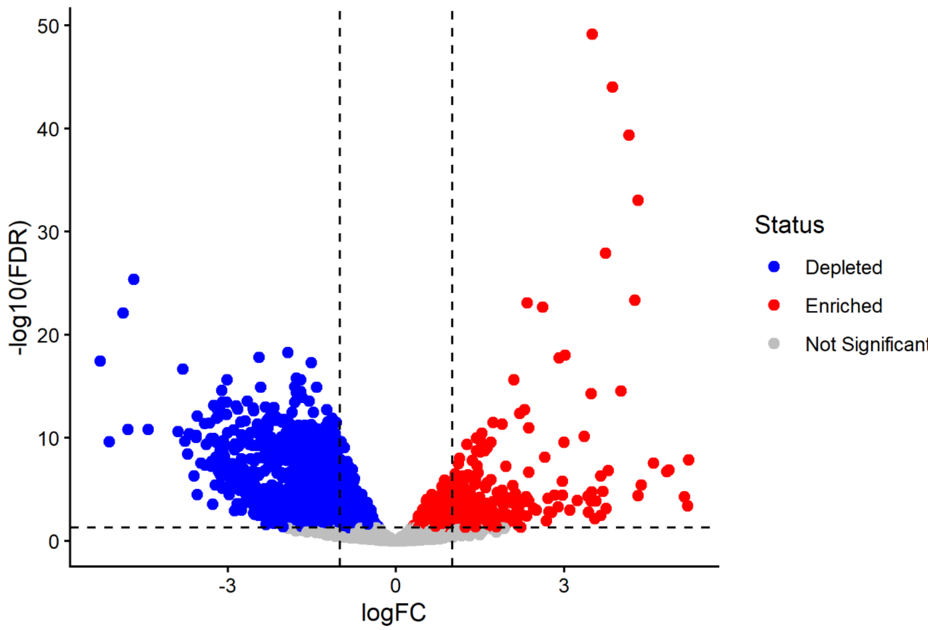

Figure 14. edgeR volcano plot, Dataset 2.

4.f Comparison across datasets

Method	Common Genera
DESeq2	80
edgeR	80
metagenomeSeq	80